

Zihao Zhao

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Education

University Hospital Aachen

Ph.D. student

Aachen, Germany

Nov.2025–Present

ShanghaiTech University

Master Student of Computer Science and Technology

GPA: 3.54/4.0

Shanghai, China

Sep.2022–Jun.2025

Harbin Institute of Technology

Bachelor of Remote Sensing Science and Technology

GPA: 90.84/100; Rank 1/25

Harbin, China

Aug.2018–Jul.2022

Research Projects

Apr. 2026 – Present: An Autonomous Agent for Clinician-driven AI Development

- We present a prototype that lets clinicians drive AI development directly. A clinician describes the task in plain language, and the system turns the description into a working pipeline, refines it through repeated experiments together with the clinician, and returns a model that meets the stated clinical objective. Under review at *Nature Communications*.

Dec. 2025 – Present: Agentic system distinguishing visually hard-to-separate diseases

- We developed a training-free multi-agent system for Pneumonia vs. Edema (Chest X-ray) and Melanoma vs. Atypical Nevus (Dermoscopy). Notably, it brings 11% point improvement in accuracy on dermoscopy data. Accepted at *MICCAI 2026*.

Feb. 2023 – Jan. 2024: Large language models as diagnostic agents

- Utilizing LLMs to interpret outputs from multiple CAD models, we improved the decision-making processes and thus proposed ChatCAD. Published in *Communications Engineering*.
- We further advanced this project by proposing ChatCAD+, which was characterized by increased reliability and capability to process inputs from multiple imaging domains. Published in *IEEE TMI*.

Sep. 2022 – May. 2025: Eye-tracking rectified computer-aided diagnosis

- For the first time, we utilized gaze data to address the issue of false-negatives in medical contrastive learning, with gaze similarity evaluation schemes tailored for various possible scenarios. Accepted for poster presentation at *AAAI 2024*.
- We extended the *AAAI* paper by developing a more advanced method to preserve spatial-temporal information during gaze similarity evaluation. This approach improves performance and validates the function of gaze data in early alignment between human knowledge and neural networks. Accepted by *IEEE TMI*.

Selected Paper

Google Scholar: <https://scholar.google.com/citations?user=Novd9cUAAAJ>

* indicates equal contribution.

- Z. Zhao, F. Hauke, J. Castilhos, S. Nebelung, D. Truhn “Can Agents Distinguish Visually Hard-to-Separate Diseases in a Zero-Shot Setting? A Pilot Study”, *29th International Conference on Medical Image Computing and Computer Assisted Intervention*, 2026. In Press.
- Z. Zhao, S. Wang, Q. Wang, and D. Shen, “Mining gaze for contrastive learning toward computer-

assisted diagnosis”, *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024, 38 (7): 7543–7551. <https://doi.org/10.1609/aaai.v38i7.28586>.

- **Z. Zhao**, S. Wang, J. Gu, Y. Zhu, L. Mei, Z. Zhuang, Z. Cui, Q. Wang, D. Shen. “ChatCAD+: Towards a Universal and Reliable Interactive CAD using LLMs”, *IEEE Transactions on Medical Imaging*, 2024, 43 (11): 3755-3766. <https://doi.org/10.1109/TMI.2024.3398350>.
- **Z. Zhao**, M. Wang, Z. Cui, S. Wang, Q. Zhou, L. Fan, Q. Wang, D. Shen. “Taming Masked Image Modeling for Chest X-ray Diagnosis by Incorporating Clinical Visual Priors”, *Information Processing in Medical Imaging*, 2025: 157-171. https://doi.org/10.1007/978-3-031-96625-5_11.
- **Z. Zhao**, Y. Liu, H. Wu, M. Wang, Y. Li, S. Wang, L. Teng, D. Liu, Z. Cui, Q. Wang, D. Shen. “CLIP in Medical Imaging: A Survey”, *Medical Image Analysis*, 2025, 102: 103551. <https://doi.org/10.1016/j.media.2025.103551>.
- S. Wang*, **Z. Zhao***, X. Ouyang, T. Liu, Q. Wang, D. Shen. “Interactive Computer-Aided Diagnosis on Medical Image using Large Language Models”, *Communications Engineering*, 2024, 3 (133). <https://doi.org/10.1038/s44172-024-00271-8>.
- S. Wang*, **Z. Zhao***, Z. Shen, B. Wang, Q. Wang, D. Shen. “Improving Self-Supervised Medical Image Pre-Training by Early Alignment with Human Eye Gaze Information”, *IEEE Transactions on Medical Imaging*, 2025. <https://doi.org/10.1109/TMI.2025.3528965>.
- S. Wang*, **Z. Zhao***, Z. Zhuang, X. Ouyang, L. Zhang, Z. Li, C. Ma, T. Liu, D. Shen, Q. Wang. “Learning Better Contrastive View from Radiologist’s Gaze”, *Pattern Recognition*, 2025, 162: 111350. <https://doi.org/10.1016/j.patcog.2025.111350>

Skills

Language: Mandarin, English (IELTS 7.5)

Programming: Python, PyTorch, C++

Awards

National Scholarship, ShanghaiTech University	2024
Academic Scholarships, ShanghaiTech University	2022-2024
Soochow Industrial Park Scholarship, Harbin Institute of Technology	2021

Activities

As a **Reviewer** for:

- IEEE Transactions on Medical Imaging
- Medical Image Analysis
- IEEE Journal of Biomedical and Health Informatics
- Pattern Recognition
- Neural Networks
- MICCAI 2025, 2026
- AAAI 2026

As a **Teaching Assistant** for:

- BME2106 Medical Big-Data and Artificial Intelligence, Spring 2024

As a **Lab Assistant** for:

- Collection of EEG and eye-tracking data
- Demonstrating the functionalities of eye-tracker during campus open days

Invited talk at:

- The 13th National Conference on Bioinformatics and Systems Biology, Haikou, China